

AI Developer Assignment

Mini AI Ticket Classifier

Goal

Create a simple Python tool that uses an open-source LLM to analyse support or incident tickets and return structured results.

Task

Your task is to build a small working prototype. It does not need to be production-ready, but it should be possible to run it and understand your approach.

1. **Use an open-source LLM model:** Use an open-source/local model or API-compatible open-source setup, for example via Ollama, Hugging Face, llama.cpp, LM Studio, or another similar option.
2. **Create 5 sample tickets:** Prepare five short example tickets. They can be technical, support-related, or incident-related. Examples: deployment failure, slow application, access problem, payment issue, monitoring alert.
3. **Design a prompt:** Write a prompt that asks the model to extract structured information from each ticket and return the result as JSON.
4. **Write a Python script:** The script should read the sample tickets, send each ticket to the selected LLM, receive the response, and print or save the resulting JSON.
5. **Add simple validation:** Validate that the model output contains all required fields and that the priority value is one of the allowed values.
6. **Add a short README:** Include instructions on how to run the solution, which tools or libraries are needed, and what you would improve with more time.

Required JSON Format

The model output for each ticket should follow this structure:

```
{
  "category": "string",
  "priority": "low | medium | high | critical",
  "summary": "string",
  "recommended_action": "string"
}
```

Validation Requirements

- The output must be valid JSON or must be converted into valid JSON by your script.
- The output must contain: category, priority, summary, recommended_action.
- The priority must be one of: low, medium, high, critical.
- If the output is invalid, the script should handle it clearly, for example by printing a useful error message.

What and how to Submit

Recommended approach: share a github/gitlab repo link. Repository should include:

- Python script or small Python project.
- The five sample tickets used for testing.
- The prompt used for the LLM.
- Example output generated by the script.
- README with setup and run instructions.

Use of AI Tools

You may use AI tools while preparing the assignment (it's highly recommended). We are interested in whether you understand the result, can run it, adjust it, and explain your decisions during the interview.

Expected Effort

Please keep the solution simple and readable. We do not expect a complex application or production-level implementation.

Notes

- We are not looking for a perfect solution. We value a practical approach, clear thinking, basic Python skills, prompt quality, and the ability to validate AI output.
- During the interview, be prepared to explain how your solution works and what you would improve next.